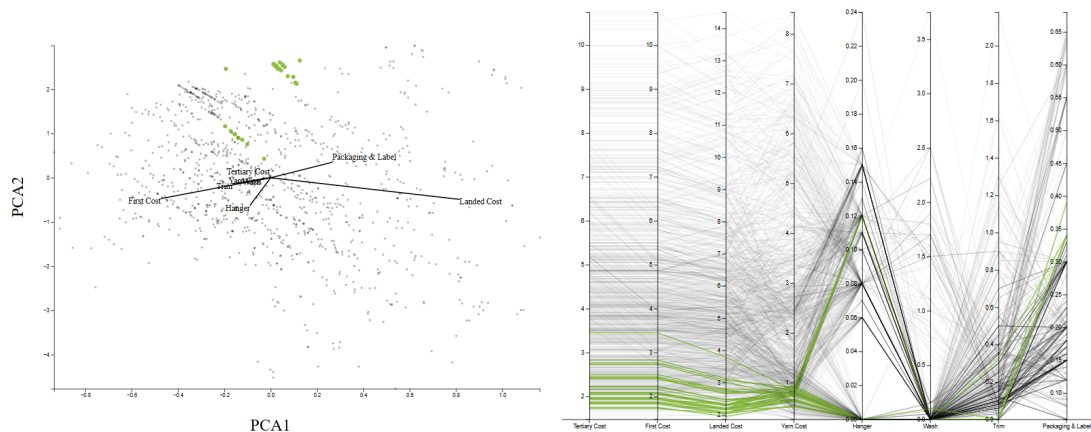


Something that I thought was really interesting about this data was that some of the item IDs yielded some very different lines on the parallel coordinates chart. For each item ID, I expected the costs to be relatively similar with also similar correlation given that most of the costs associated with making the same item are pretty similar. However, I found that some item IDs like 119 and 144 tend to not follow that correlation pattern. Most follow a more correlated path with each other. But what still surprised me at times was how the first cost for the same item could be so different for different vendors in the same item ID.

One other thing is that with parallel coordinates and PCA, we are able to make a relatively educated guess at what is going on. Take this for example of Vendor M



We could possibly tell with this in conjunction with the PCA plot, we can see that hanger and Packaging and labeling were two key differentiators as confirmed by the parallel coordinates chart. But if we want to know whether that means that vendor M has very competitive prices, we would have to see whether Vendor M just produces very simplistic items so the prices M is producing is less for first and landed costs, or that they managed to bring down costs enough to beat most of the competition.

For the K-means cluster, It seemed to mostly cluster really well based on the things that correlated the most like tertiary cost, first cost, and landed cost. The interesting thing is that if those three costs were low to begin with, they would ultimately stay low for everything else minus the packaging labeling and there was a split where half went to hanger and half went to wash, meaning that in the low cost sector, there are two cost types. Possibly it could be where one of types of items is just easy to mass produce and mass ship without much effort items and especially since the lines are thickest towards the bottom, it seems to be the case that most items fall in this category. The other low cost item has big hanger cost and distributed trim and packaging and labeling, but no wash. We could surmise they might be more simplistic version of complicated clothing. And the other clusters mostly deal with the higher quality and higher service goods. While the clusters don't tell a complete picture without the vendor part of the bar graph + parallel coordinates to tell the vendor tendencies, but it starts to paint a more complete picture of what is going on.